AI-driven Dynamic Weighted Edge Computing Scheduler

GROUP B6

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Problem Statement

- MEC servers have limited computing power and energy resources.
- Tasks arrive randomly and unpredictably, making scheduling complex.
- Efficient task scheduling is critical to ensure timely task completion.
- Poor scheduling leads to lower system performance and reduced user satisfaction.
- **Main Challenge:**
  - Designing a scheduler that balances task completion, deadlines, and energy efficiency

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Objectives

- Develop an intelligent scheduler for heterogeneous servers.
- Implement Lightweight Q-learning to dynamically adjust scheduling weights.
- Optimize for:
  - Task completion before deadlines.
  - Reduce total energy consumption.
- Compare performance against traditional scheduling methods.

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Literature Review

S.No.	Name	Publisher|Year	Inference	Cons.
1	Research on Task Scheduling Methods in 5G Mobile Edge Computing Environments	IEEE-2024	Improved task completion vs FCFS, EDF, SJF by using weighted factors	Fixed weights; not adaptive
2	Resource Calendaring for Mobile Edge Computing in 5G Networks	IEEE-2021	Efficient resource utilization, Reduced latency & Better load balancing	Extra overhead to maintain calendar

Literature Review

S.No.	Name	Publisher|Year	Inference	Cons.
3	Resource Adaptive Automated Task Scheduling Using Deep Deterministic Policy Gradient in Fog Computing	IEEE-2024	Reduced latency & energy with adaptive RL	High training complexity
4	Faashouse: Sustainable Serverless Edge Computing Through Energy-Aware Resource Scheduling	IEEE-2024	Improved utilization & sustainability in serverless MEC	Restricted to serverless; not general MEC

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Research Gap & Novelty

Research Gap

- FCFS, EDF, SJF → simple but inefficient in MEC.
- DWECS improves scheduling but uses static weights.
- Advanced RL methods exist but are complex and resource-heavy.

Novelty

- Dynamic weight adjustment using lightweight Q-learning.
- Balances task completion & energy adaptively.
- Lightweight design → suitable for real-time MEC.
- Provides fair comparison with FCFS, EDF, SJF, and DWECS.
- Adaptive to dynamic task arrivals and heterogeneous MEC servers.

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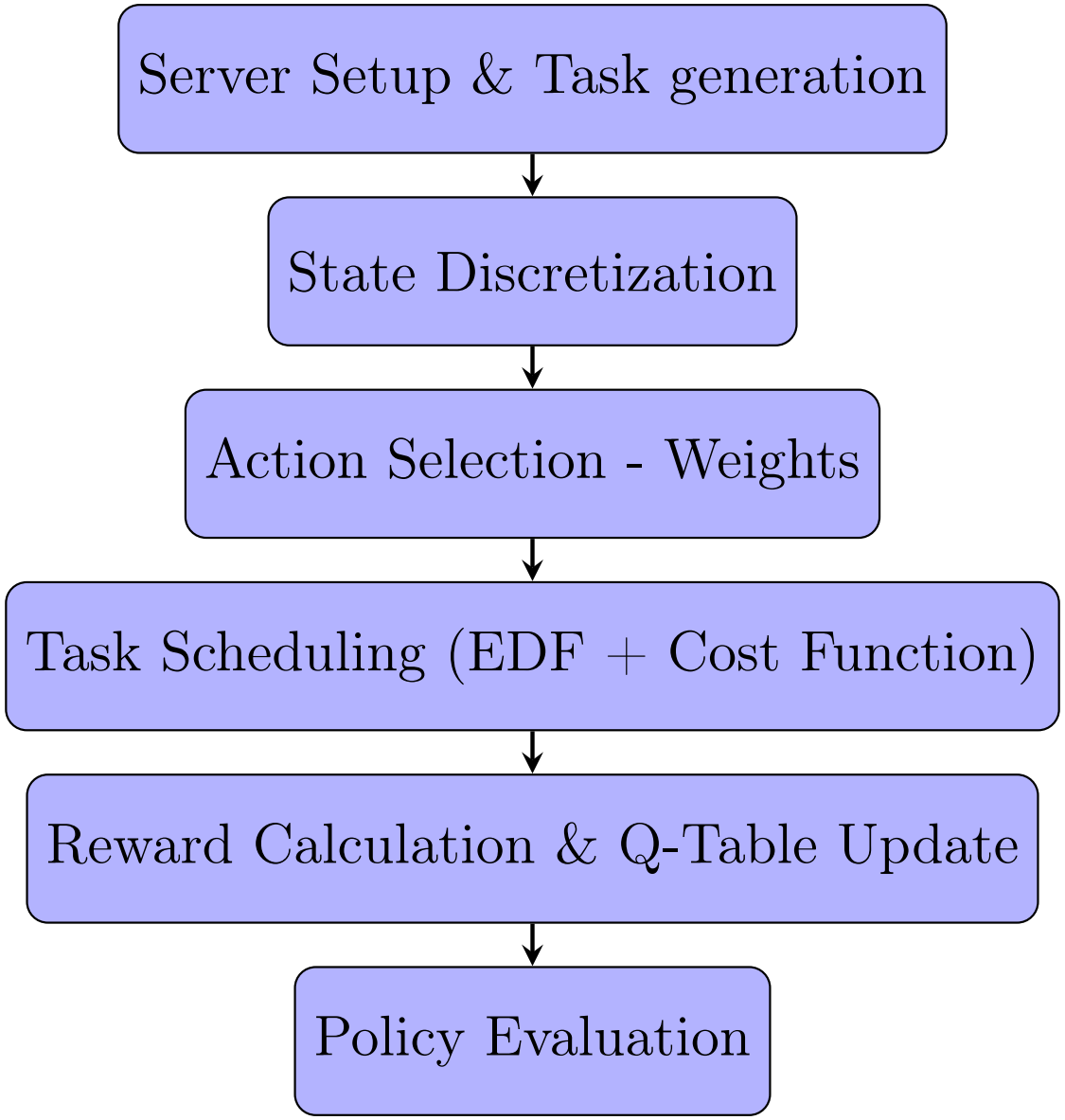

Methodology

Server Setup & Task Generation

- Create 3 heterogeneous servers. And generate random tasks with varying computation loads and deadlines to simulate a real-time MEC environment.

State Discretization

- Represent the system state using average task urgency and average task load.
- Reduces the state space for lightweight Q-learning.

Action Selection - Weights

- Choose scheduling weights for time vs energy trade-off.
- Use epsilon-greedy policy: explore randomly sometimes, otherwise exploit the best-known action.

Methodology

Task Scheduling (EDF + Cost Function)

- Apply Earliest Deadline First (EDF) to order tasks.
- Assign tasks to servers minimizing a weighted cost function (time + energy).

Reward Calculation & Q-Table Update

- Reward = tasks completed − penalty × energy consumed.
- Update Q-table using lightweight Q-learning for adaptive scheduling.

Policy Evaluation

- Measure task completion rate, energy consumption, and execution efficiency.
- Compare the results.

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Implementation

Server details

- Server 1: GHz = 2.0, Power = 10 W
- Server 2: GHz = 3.0, Power = 15 W
- Server 3: GHz = 1.5, Power = 7.5 W

Task Generation

- Batch size: 10 tasks per batch
- Load: 2–15 billion cycles
- Deadline: 2000–20000 ms
- Arrival: Sequential

Implementation

State Space

Compute avg_deadline → Urgency_Bin: 1 if <6000 ms, else 0

Compute avg_load → Load_Bin: int(avg_load // 3e9)

Load Bin: 0 (light), 1 (medium), 2 (heavy) based on average task load.

Action Space

Predefined weight combinations for cost function

Each action decides how scheduler balances completion vs energy.

Implementation

EDF Ordering: Tasks sorted by prioritizing the earliest deadline.

Execution Time

$$T_{\text{exec}} = \frac{\text{Load (cycles)}}{\text{GHz} \times 10^9} \times 1000 \tag{1}$$

Energy Formula

$$E = \frac{T_{\text{exec}}}{1000} \times P \tag{2}$$

Server Availability

$$FinishTime = AvailableTime + ExecutionTime \tag{3}$$

Implementation

Normalize time and energy

Cost function

$$\text{Cost} = w_{\text{time}} \cdot \text{NormTime} + w_{\text{energy}} \cdot \text{NormEnergy} \tag{4}$$

- This ensures the task is assigned to the server that best balances execution time and energy according to the current weights.
- Logging - a record of each task is saved for reporting or debugging.

Outputs

- Completion Status
- Energy Consumed
- Server assignments

Implementation

Training: 10000 episodes run.

- Lightweight Q-learning with small Q-table
- Epsilon-greedy policy for exploration vs exploitation

Reward Function

$$\text{Reward} = \text{Tasks Completed} - 0.002 \times \text{Energy Consumed (J)} \tag{5}$$

Q-Table Update

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right] \tag{6}$$

Evaluation: Choose the task based on exploitation.

- Learned policy compared against EDF, FCFS, SJF.
- Measures: Task completion rate & Energy consumption

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Results & Discussion

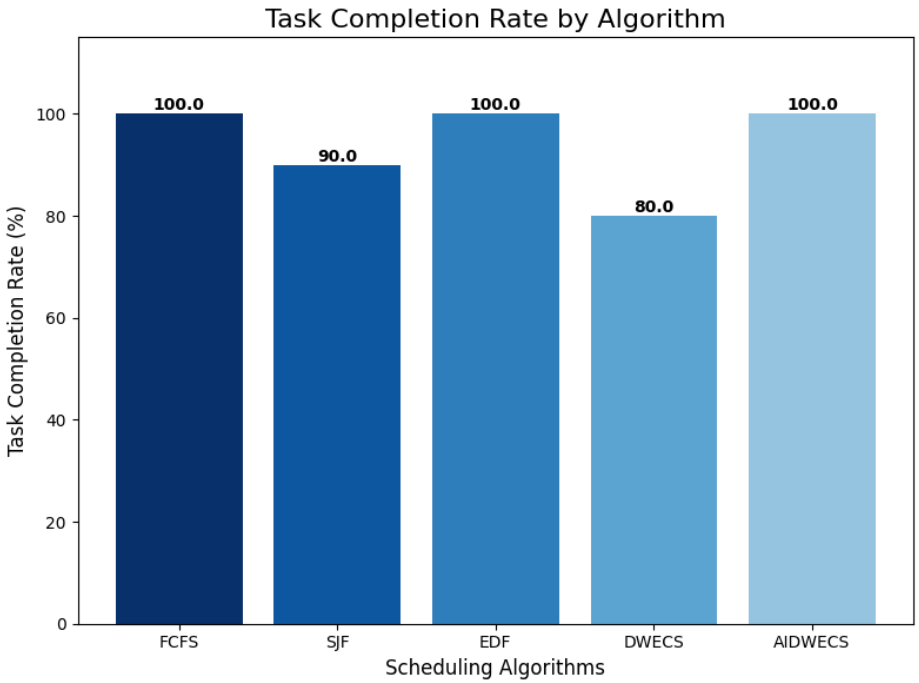
Tabela 1: 1st Task Batch for Evaluation

ID	Load (Cycles)	Deadline (ms)	Arrival Time (ms)
1	3×10^9	2000	0
2	10×10^9	10000	500
3	2×10^9	3000	1000
4	4×10^9	4500	1500
5	12×10^9	15000	2000
6	3.5×10^9	7000	3000
7	5×10^9	7500	4000
8	2.5×10^9	8000	5000
9	8×10^9	14000	5500
10	6×10^9	13000	6000

Results & Discussions

Tabela 2: Comparison of Task Completion Rate

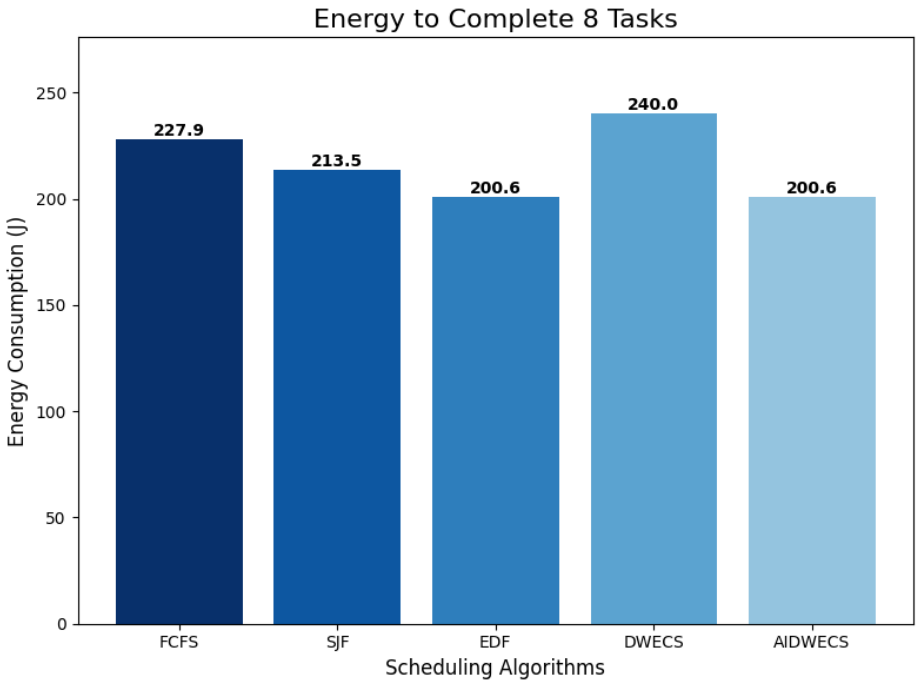
Scheduler	Task Completion (%)
FCFS	100.0
SJF	90.0
EDF	100.0
DWECS	80.0
AIDWECS	100.0



Results & Discussion

Tabela 3: Comparison of Energy Consumption for 8 Tasks

Scheduler	Energy Consumption (J)
FCFS	227.9
SJF	213.5
EDF	200.6
DWECS	240
AIDWECS	240.6



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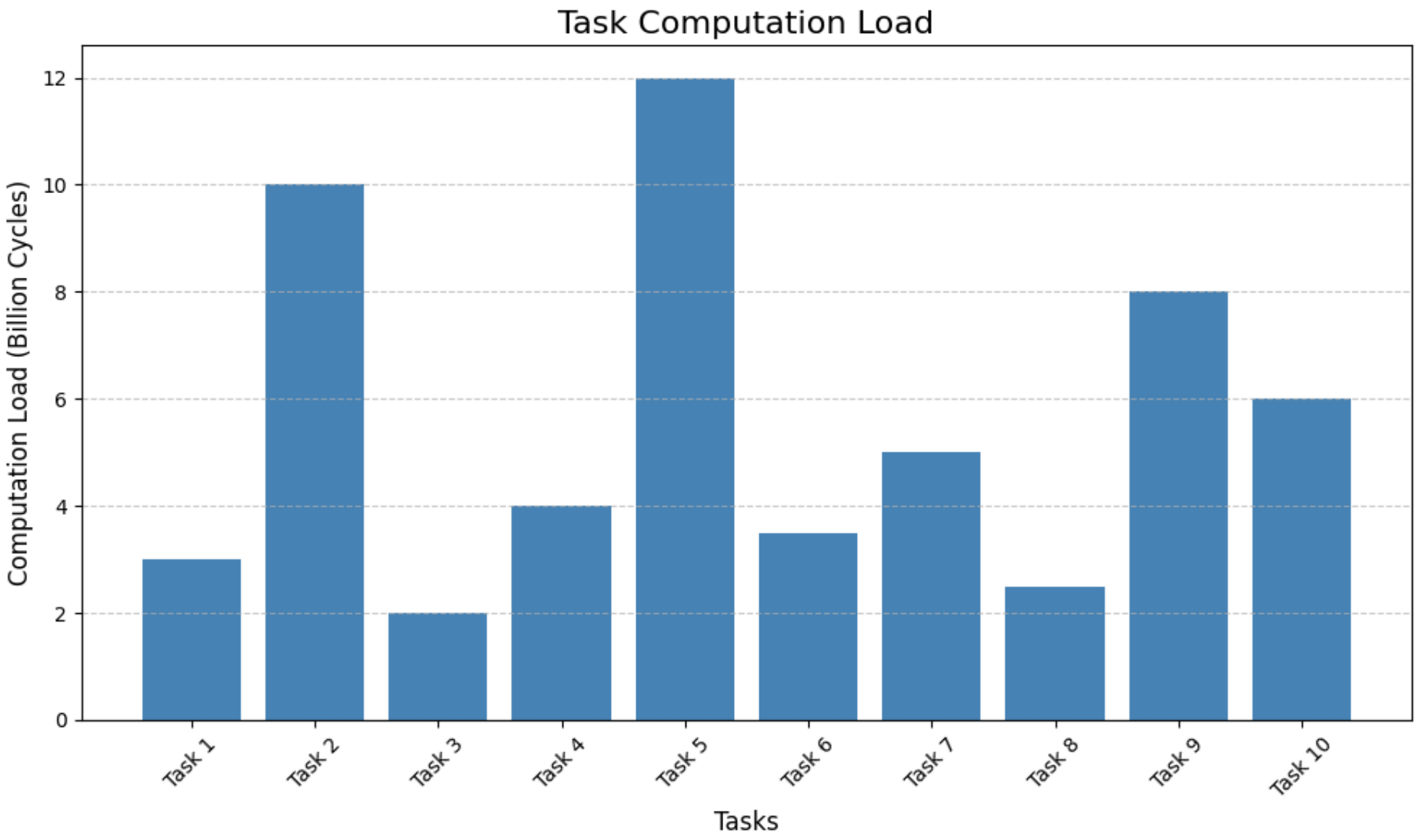


Figura 3: Task Computation Load

Results & Discussion

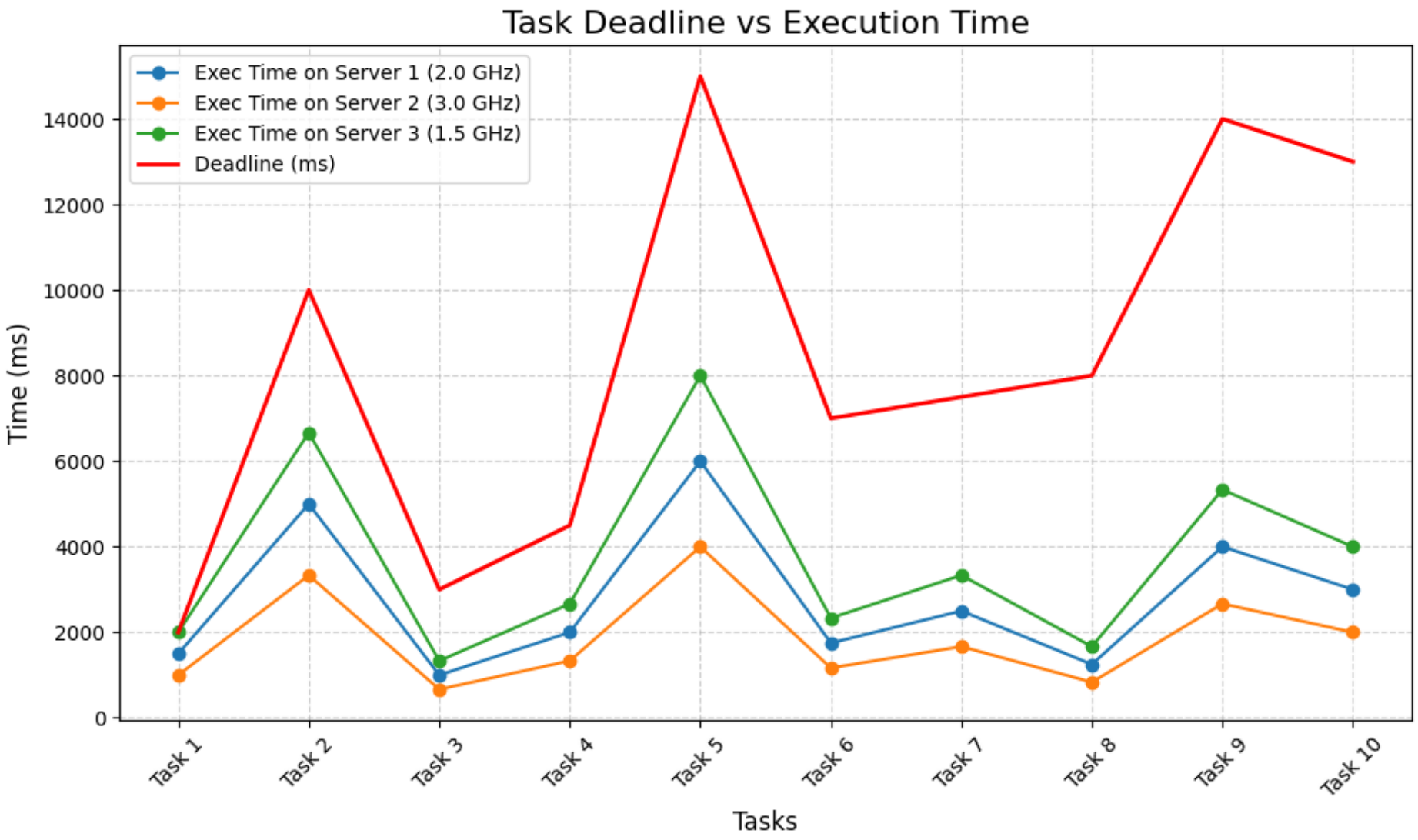


Figura 4: Task Deadline Vs. Execution Time

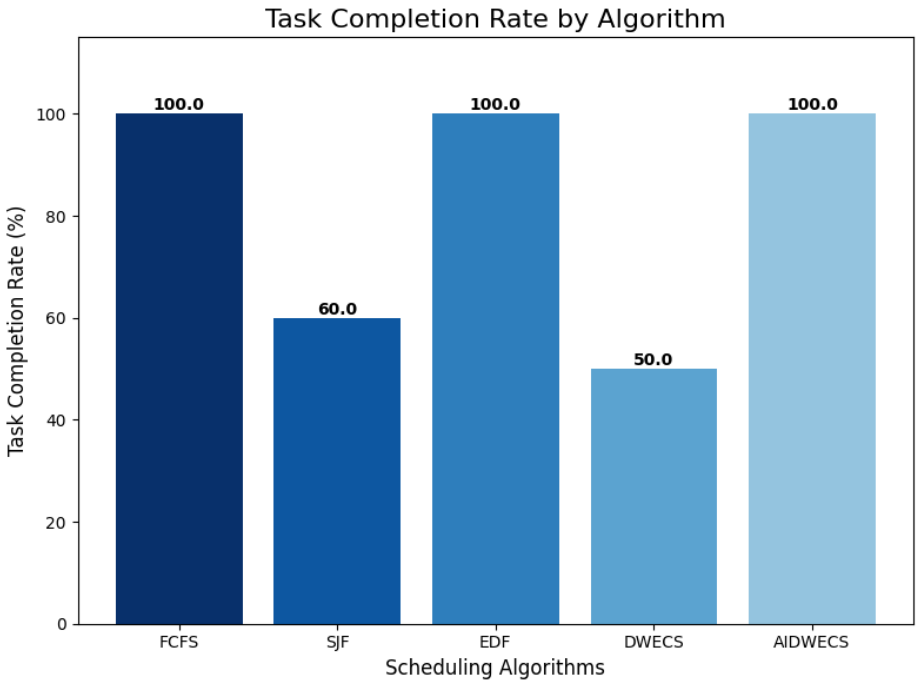
Tabela 4: 2nd Task Batch for Evaluation

ID	Load (Cycles)	Deadline (ms)	Arrival Time (ms)
1	4×10^9	2500	0
2	8×10^9	9000	0
3	6×10^9	5000	0
4	3×10^9	2800	200
5	5×10^9	6000	500
6	10×10^9	12000	2000
7	2×10^9	8000	3500
8	7×10^9	11000	4500
9	3.5×10^9	9500	6000
10	9×10^9	16000	7000

Results & Discussions

Tabela 5: Comparison of Task Completion Rate

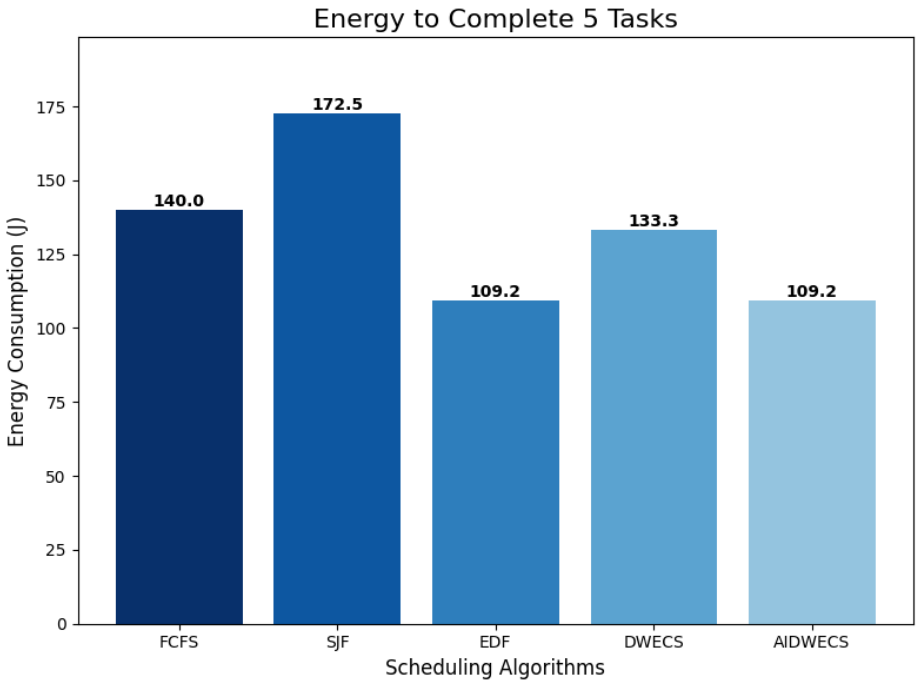
Scheduler	Task Completion (%)
FCFS	100.0
SJF	60.0
EDF	100.0
DWECS	50.0
AIDWECS	100.0



Results & Discussion

Tabela 6: Comparison of Energy Consumption for 5 Tasks

Scheduler	Energy Consumption (J)
FCFS	140.0
SJF	172.5
EDF	109.2
DWECS	133.3
AIDWECS	109.2



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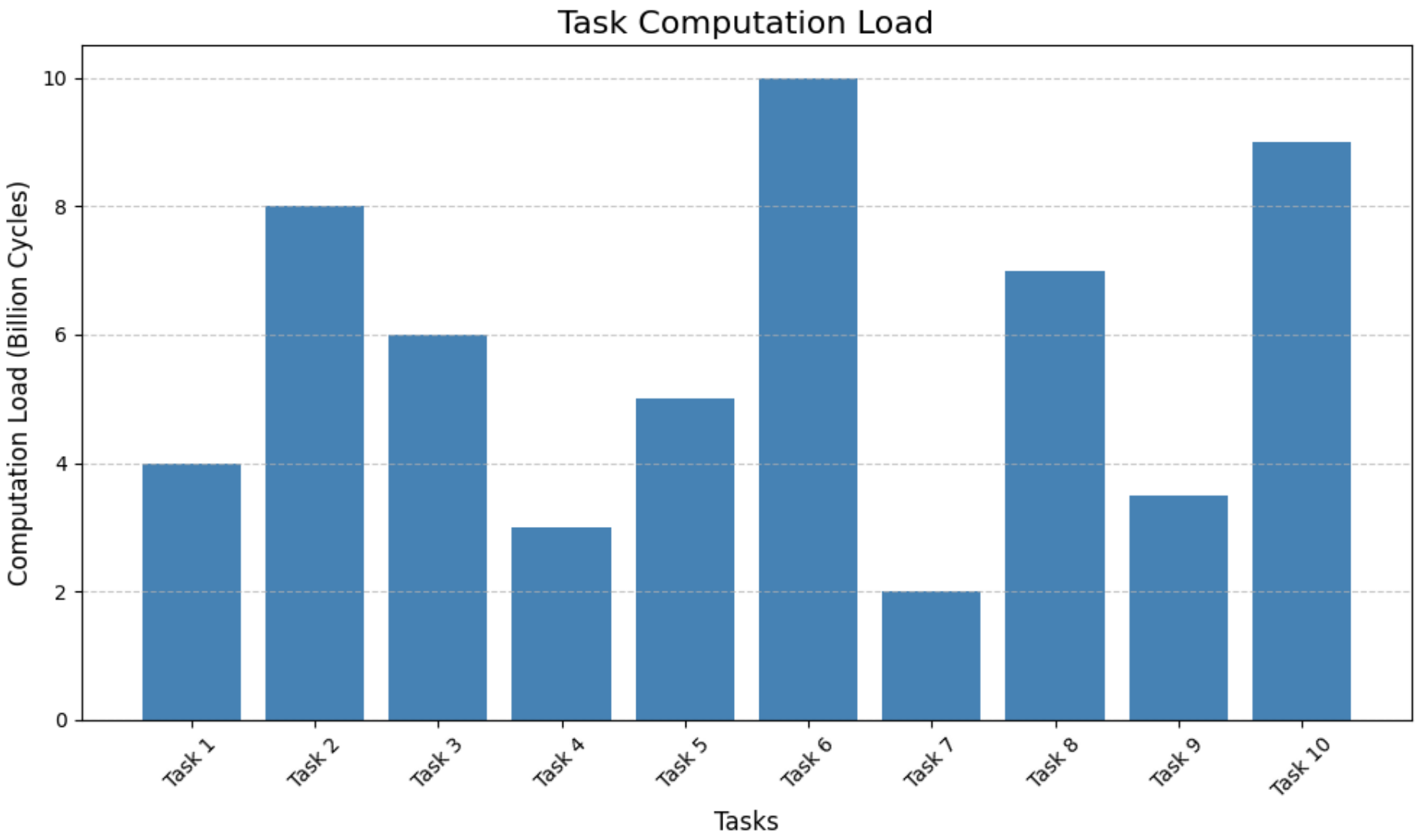

Figura 7: Task Computation Load

Results & Discussion

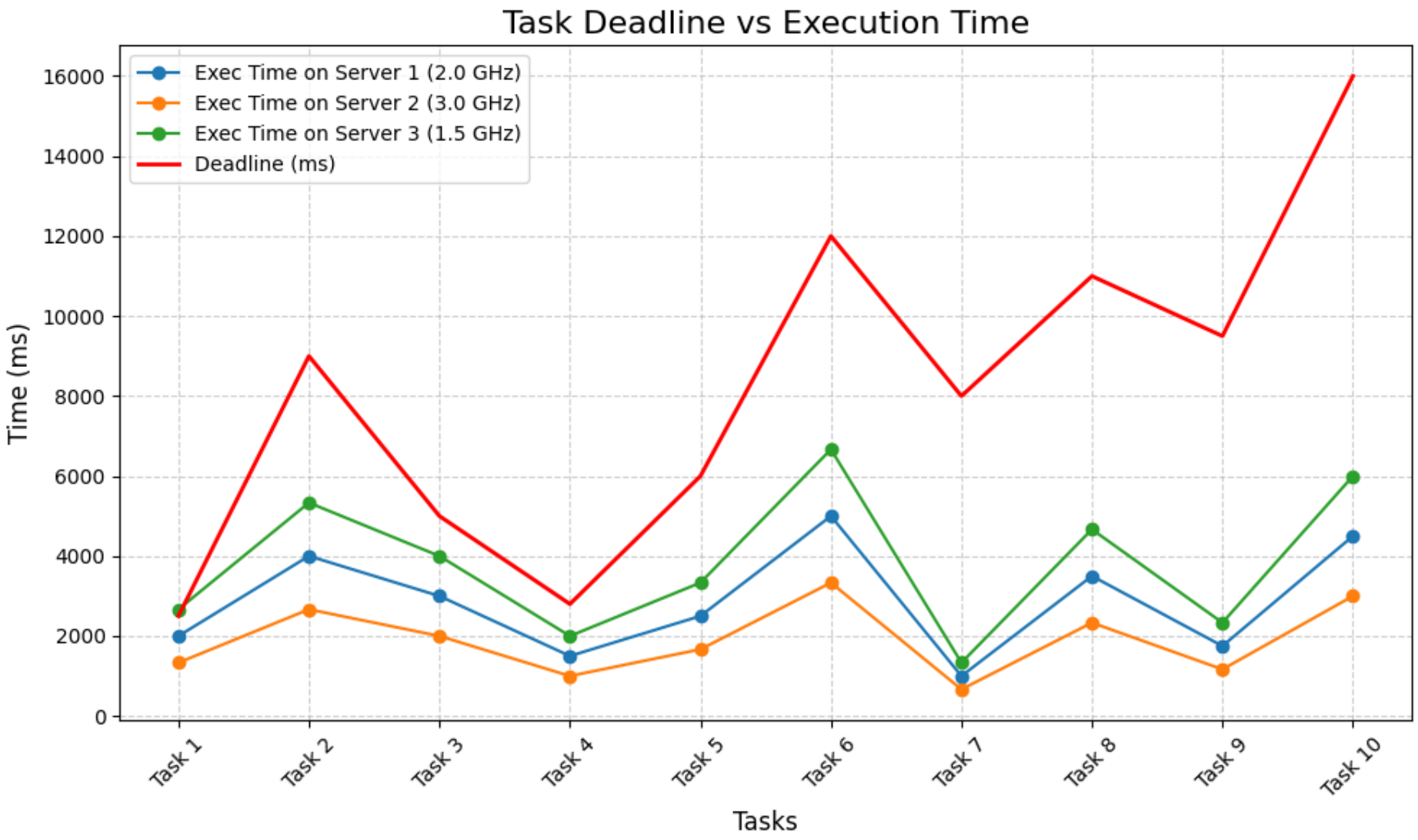

Figura 8: Task Deadline Vs. Execution Time

Results & Discussion

Task Scheduling Performance Comparison

- Schedulers Compared: EDF, FCFS, SJF, DWECS, AIDWECS.
- Metrics Evaluated: Task Completion Rate(%), Total Energy Consumption (J) & Task Deadline vs. Execution Time.

Key Observations:

- EDF and AIDWECS achieve the highest completion rates and lowest energy consumption for the tested task batch.
- FCFS, SJF & DWECS performs comparitively worse.
- The scheduling strategies EDF, AIDWECS improves the task success.
- But since both EDF and AIDWECS show same results, we will compare them again.

Results & Discussion

Tabela 7: Comparing EDF & AIDWECS Stress Test

Scheduler	Server	Deadline	Exec Time	Avl At	Finish Time	Violation
EDF	Server 2	12000	1333.33	13000	14333.33	2333.33
AIDWECS	Server 1	12000	2000	11000	13000	1000

- In a high-congestion stress test, AIDWECS mitigated the deadline violation to 1,000 ms, while the static EDF failed with a 2,333 ms delay.
- Conclusion: This proves AIDWECS provides 57% better system reliability by adapting to resource congestion, a critical capability EDF lacks.

Tabela 8: Worst-Case Time Complexity Comparison (N: Tasks, M: Servers)

Algorithm	**Time Complexity**	**Validation/Justification**
FCFS, SJF, EDF	$O(N \log N + N \cdot M)$	Complexity is dominated by the initial sort of N tasks ($O(N \log N)$), plus the cost of checking M servers for each of the N tasks ($O(N \cdot M)$).
DWECS	$O(N \log N + N \cdot M)$	Requires $O(N \cdot M)$ to find max values for cost normalization. Uses $O(N \log N)$ for the initial sort. Main scheduling loop is $O(N \cdot M)$.
AIDWECS	$O(N \log N + N \cdot M)$	Pre-scheduling includes an $O(1)$ Q-table lookup. Complexity is dominated by sorting ($O(N \log N)$) and the necessity of $O(N \cdot M)$ pre-computation and server selection.

Tabela 9: Worst-Case Space Complexity Comparison (N: Tasks, M: Servers)

Component	**Space Complexity**	**Validation/Justification**
All Schedulers	$O(N)$	The primary memory usage scales with the number of tasks being stored (e.g., deep copy of tasks and output logs). Server state is a constant $O(M)$.
AIDWECS (Q-Table)	$O(1)$	The state space is fixed by the small, constant number of discretization bins (e.g., Urgency $\times$ Load). Since the number of states and actions is fixed, the storage is constant.

Tabela 10: Per-Task Effort (Constant Factor) Analysis in $O(N \cdot M)$ Loop

Algorithm	**Complexity per Task**	**Constant Factor Analysis (Decision)**
FCFS, SJF, EDF	$O(M)$	**Lowest Constant Factor**. Decision is a simple comparison of finish times across M servers, requiring minimal arithmetic per server.
DWECS	$O(M)$	**Higher Constant Factor**. Involves calculating multiple metrics (time, energy, tardiness), normalizing them (division), and then summing the weighted values for each of the M servers.
AIDWECS	$O(M)$	**Highest Constant Factor**. Performs the same complex multi-metric calculation and normalization as DWECS, plus an $O(1)$ Q-table lookup for weights before the loop, leading to the highest operational cost per task-server pair.

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Conclusion

- Best Performing Schedulers: EDF and AIDWECS consistently achieve the highest task completion rates and lower energy consumptions.
- Energy Efficiency: AIDWECS & EDF shows the least energy consumption to complete a fixed number of tasks.
- Effectiveness of Intelligent Scheduling:
  - Though AIDWECS & EDF gives the same results, AIDWECS performs well by demonstrating superior adaptive intelligence in high-congestion states, mitigating critical deadline violations by 57% where EDF's rigid policy failed.
- Using deadline-aware and cost-aware schedulers improves system performance with the least possible energy consumption, making them ideal for heterogeneous edge environments.

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THANK YOU!!

Q & A